

Latency

In the future generation wireless systems, the latency will become a key performance metric. Why it is becoming important?



Many applications will require high responsiveness in the interaction with the network



Many applications will require high throughputs at the end users



1)

The end users may have high mobility during their connection



Many applications will use high data processing and computation

Multi-path

What is important to be modeled in the multipath happening in a wireless communication, to use it constructively?



The attenuations and the delays of the different paths



The attenuation of the line of sight path



The delay of the line of sight path



Only the attenuation and delay of the line of sight path

PON

Which are the benefits of the Passive Optical Networks compared to the Active ones?



The cost of the infrastructure is lower



It is possible to reach longer distances only with fibers



The traffic management is easier



We can avoid to use Optical Network Units at the user side

VDSL

Which are the two improvements enabling the VDSL?



The vectoring of the signals in the copper cables and the placement of the DSLAM close to the end user



The vectoring of the signals in the fibers cables that arrive till the end users



The use of high frequencies in the copper cable band and the full cancellation of the cross-talk noise



The use of high frequencies in the fiber optic cables and the vectoring of these signals

ADSL success

One of the key element of the ADSL success is related to:



The use of the same copper cable of the old telephone network



The use of the same frequency band of the old telephone network



The use of new fiber cables in place of the old telephone ones



The fact that no new network elements are needed at the Central Office side

LA C

How the problem arising in a PON when a number of Optical Network Units (ONU) try to transmit contemporarily is solved?

Select one:

- ☐ a. The problem cannot be solved, but the ONUs are able to retransmit after a collision of their packets.
- ☐ b. The power emitted by all the ONUs is regulated in order to have the same value.
- ☐ c. The OLT controls the transmissions of the ONUs by giving the right scheduling to avoid collisions.
- ☐ d. The ONUs buffer their data until they find the fiber free for their transmission

Cross_talk

What is the cross-talk noise on ADSL?



The coupling of signals from a circuit to another giving rise to undesired interference



A noise that is generated at a generic receiver within a fiber cable



A noise that is generated at a generic receiver within a copper cable



The interference of a signal in a frequency band to signals in other frequency bands of the ADSL

EDGE

What is the EDGE of a network and which is its role?



That part of the network in the between of the access and the core with some intelligent functions



The very peripheral part of the network where all the applications are



The very peripheral part of the network where only simple functions are implemented



The very core part of a network were powerful network functions are implemented

Evolution current access network

The evolution of the current access network aims at:



Employing fiber cables instead of copper ones



Using only wireless access given its flexibility and low cost



Using in a more capillary way the already existing infrastructure based on copper cables



Providing an ADSL access to all the users

FTTX

Which one of these configurations is the best in terms of expected bit rate?



FTTB - to the building



FTTE- to the exchange



FTTC -to the curb



ADSL

Topology

Which are the most used topologies for networks in the access part?



Stars and rings



Rings



Bus



Mesh and bus

Risposta: A

How the Wavelength Division Multiplexing (WDM) is a fiber-optic transmission for a PON is used?

Select one:

- ☐ a. If we have 10 different wavelenghts we can have 10 different colors contemporalry in the same fiber.
- ☐ b. The same Optical Network Unit can trasmit different pakets on different wavelenghts.
- ☐ c. One wavelenght is used for the downstream and one for the upstream.
- ☐ d. If we have 10 different wavelenghts we can have 10 different fibers contemporalry in the same cable.

Risposta: B

Which are the benefits of the Passive Optical Networks compared to the Active Optical ones?

Select one:

- ☐ a.
We can avoid to use Optical Network Units at the user side
- ☐ b. The cost of the infrastructure is lower
- ☐ c.
It is possible to reach longer distances only with fibers
- ☒ d.
The traffic management is easier thanks to the use of Ethernet based approches

packet auth

In order to authenticate the entire IP packet, IPSec peers must use:



AH in tunnel mode



AH in transport mode



ESP with authentication in transport mode



it is not possible to authenticate all the header fields since some of them change value while transiting in the network

Outbound IPSec

With reference to the outbound processing performed by an IPSec peer:



the first database that is checked is the SPD in order to determine if the current packet requires security service



the first database that is checked is the SAD in order to determine the set of security algorithms to use to handle the current packet



the SPI is used to select the proper SA to use to process the incoming packet



the SPD and SAD are inspected only if the incoming packet requires IPSec processing

ssh port forwarding

What is the command to create a local port forwarding of local port 8080:

☐

ssh - NL local_p:host_ip_addr:remote_p user@tunnel_ip -p 8080

☐

ssh - NL remote_p:host_ip_addr:8080 user@tunnel_ip

☒

ssh - NL 8080:host_ip_addr:remote_p user@tunnel_ip

☐

ssh - NL 8080:host_ip_addr:8080 user@tunnel_ip

SA negotiation

Let us consider a TCP connection steered over an IPSec tunnel, then:

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SAs might be re-negotiated while the flow is alive

☐

before negotiating new SAs, the security gateways need to wait for the TCP connection tear down

☐

SAs are re-negotiated only in case the IPSec sequence numbers overcome the maximum value

☐

SAs do not need to be negotiated if these have been statically configured